

Article

Metadata-Grounded Large Language Models for Converting Hospital Statistical Requests into Clinical Data Warehouse Queries

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Featured Application

The framework converts free-text statistical requests from clinical departments into executable, provenance-carrying SQL over a hospital clinical data warehouse, so that the departmental performance, quality-control and regulatory indicators on which hospital management depends are delivered with the counting criterion recorded beside the number rather than left in the query. It runs on premises, so no patient data leaves the institution, and it refers a contested criterion back to the indicator owner instead of resolving it silently.

Abstract

Hospital reporting depends on institution-specific statistical criteria—which timestamp a case is counted by, which flag defines an indicator—that carry no cross-site vocabulary, so requests from clinical departments reach a data engineer before they reach the database. We convert such requests into executable Presto SQL over a three-layer clinical data warehouse, grounding generation in a metadata knowledge base and verifying each candidate through syntax, field, criterion and result checks. On 398 business terms from 30 reporting tasks, mapping accuracy was 52.3% against 32.9% for the rule-based matcher in production ($p < 0.001$). A factorial experiment over 24 tasks, 8 models and 5 prompting strategies produced 960 attempts: 791 yielded SQL, 274 with a field or criterion error. The on-premises qwen2.5:7b model achieved the highest effective SQL rate, 71.7% (95% CI 63.0–79.0), against 60.8% for gpt-4o, because it hallucinated less, not because it generated more. Wrong counting criteria formed the largest error class at 34.7%, a failure that yields an executable query and a wrong number, and that generic schema-linking taxonomies do not capture. Because that failure silently changes a reported indicator’s denominator, we state which performance-management and quality-control uses the measured reliability supports, and which risk stays with the institution.

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1. Introduction

Hospital operational reporting is essential for quality control, performance management, and clinical research, yet it faces a persistent bottleneck in data retrieval. In our institution, statistical requests submitted by clinical departments, quality-management offices, and researchers are queued to a small team of data engineers; over the 2023–2024

period, [XX]% of requests waited more than five working days for a first result, and [XX]% required at least one round of criterion clarification after delivery. These delays defer management decisions and, for research requests, extend study timelines and costs, ultimately limiting how often clinical questions are asked of data that the institution already owns.

Delay is not the whole cost. Hospital management runs on indicators that are compared: one department against another in performance review, one month against the last in quality control, and one institution against its peers in regulatory returns. A comparison carries information only if the counting criterion behind each number is the same, and when every report is written by hand that criterion lives in the query rather than in any catalog the institution can inspect. Two analysts asked in different quarters for the volume of emergency surgery may filter on different flags and each be able to defend the choice; what management then reads as a trend is an artifact of who wrote the query. The failure is quiet, because a query built on the wrong criterion still runs and still returns a plausible number. What a reporting pipeline owes a hospital is therefore not only the number but the criterion it was counted by, in a form the owner of the indicator can review—and, where the criterion is genuinely contested, a way of saying so instead of choosing one silently.

Clinical data warehouses (CDWs) exist to relieve exactly this pressure. Ours follows a three-layer design—data lake (DL), data center (DC), and master data repository (MDR)—in which surgery and anesthesia, blood transfusion, laboratory, pathology, and billing systems are standardized into subject-oriented models queried through Presto [1]. That removes much of the join complexity; internationally the same role is played by shared common data models such as the Observational Medical Outcomes Partnership model [2], though our reporting is defined against institution-specific indicators rather than a shared vocabulary. However, transforming a free-text statistical request into a warehouse query remains difficult, because the request encodes *statistical criteria* rather than data structures. Room-in time, surgery-completion time, discharge time, and application time are not interchangeable for counting; “elective surgery,” “unplanned reoperation,” and “complication” are institution-specific definitions whose counting object, time criterion, and filters must be recovered before any SQL can be written.

Querying structured clinical data directly from natural language has been explored along three lines. The text-to-SQL line of work has progressed from sequence-to-sequence parsers to schema-linking, constrained-decoding, and decomposed in-context architectures evaluated on public benchmarks [3–10], and has been carried into the clinical setting through EHR-specific datasets, generation models trained on them, and reliability-oriented shared tasks that reward abstention on unanswerable questions [11–15]. The evaluation framework used here is adopted from one of those studies, which converted clinical-trial eligibility criteria into Observational Medical Outcomes Partnership Common Data Model queries and profiled hallucination across eight models under five prompting strategies [16]: the eight-model cloud-versus-local comparison, the five prompting strategies, the split into development and validation task sets, and the five-level severity ordering of hallucination categories are theirs, and we reuse them so that our numbers can be read beside theirs. What differs is the object being queried. Their criteria are eligibility rules expressed against a shared common data model, where a concept either exists in the vocabulary or does not; ours are institution-specific counting criteria, where the field exists, the query executes, and the number is still wrong. That difference is what Category B of our taxonomy measures, and it has no counterpart in a shared-vocabulary setting. A separate evaluation of six recent models on MIMIC-III and TREQS reports the same accuracy–cost tension and names retrieval augmentation and terminology integration as the remedies worth trying [17]; the system reported here implements both, and quantifies what each contributes. A parallel line converts free-text eligibility criteria into executable

cohort queries, from rule- and parser-based systems to modular query generators [18–20]. Metadata enrichment with task decomposition has been applied to clinical registry querying and reported substantial gains [21]. Despite these advances, the automated generation of warehouse queries from statistical requests faces a critical reliability challenge: LLMs frequently emit nonexistent column and table identifiers, and—more insidiously for hospital reporting—assign a plausible but wrong counting criterion, a failure that produces an executable query and a wrong number.

Recent studies have shown that LLMs, while powerful for natural language understanding, exhibit substantial error rates when mapping domain terms onto controlled vocabularies and schemas, and that these rates vary widely with model and prompt [22–24]. This reliability gap poses a significant barrier to clinical deployment, because an incorrect criterion mapping does not fail loudly—it silently changes the denominator of a quality indicator that may be reported to regulators. Furthermore, hospital deployment adds a constraint that benchmark studies do not face: patient data may not leave the institution, so cloud-hosted models can only be used on de-identified metadata, and the choice between cloud and on-premises models materially affects both accuracy and cost. Systematic comparisons across models under this constraint remain limited.

This study addresses these challenges through a two-pronged approach. First, we present an end-to-end automated system that transforms free-text hospital statistical requests into Presto SQL queries compliant with our three-layer CDW, validated with real patient data from West China Hospital, Sichuan University. Second, we conduct a comprehensive evaluation of hallucination patterns across 8 LLMs (both cloud-based and on-premises) using SynHDW, a de-identified synthetic mirror of the warehouse, systematically analyzing how model selection and prompt engineering affect reliability. Our findings reveal that while LLMs can accelerate query generation from days to minutes, hallucination rates of 20%–47% highlight the need for careful model selection and validation strategies. Surprisingly, our evaluation demonstrated that model scale does not necessarily correlate with performance, with smaller on-premises models outperforming larger commercial alternatives in terms of effective SQL generation. By quantifying these trade-offs and identifying optimal model–prompt combinations for different reporting scenarios, this work advances the development of trustworthy and cost-effective AI systems for hospital data operations.

2. Materials and Methods

2.1. Study Design

We conducted a two-phase investigation to develop and validate an automated system for transforming hospital statistical requests into Presto SQL queries over a three-layer CDW, followed by systematic analysis of LLM hallucination patterns. The integrated workflow enables both functional validation and reliability assessment of LLM-generated queries in real-world reporting scenarios.

2.2. Data Sources and Study Population

2.2.1. Statistical Request Dataset

We extracted statistical requests from the reporting ticket system of West China Hospital, Sichuan University (accessed on 27 July 2026), retaining only tickets that had been closed with an engineer-authored, criterion-confirmed reference SQL query. We utilized two distinct datasets: (1) Development Dataset and (2) Validation Dataset.

First, in the Development Dataset, 30 reporting tasks (10 each from the surgery and anesthesia, blood transfusion, and laboratory domains) were used exclusively for metadata dictionary construction and prompt optimization. From these tasks, we extracted 412

unique business terms for comparative analysis of mapping approaches between the LLM and RBFM.

Second, in the Validation Dataset, 7 high-usage reporting tasks were selected based on request frequency over the preceding 24 months (RPT-001, monthly elective surgery volume by department; RPT-002, unplanned reoperation rate; RPT-003, postanesthesia care unit [PACU] throughput and delayed-discharge rate; RPT-004, monthly blood transfusion volume and transfusion-reaction rate; RPT-005, critical laboratory value notification timeliness; RPT-006, surgical site infection surveillance denominator; RPT-007, day-surgery share of total operative volume) for comprehensive system evaluation.

2.2.2. Clinical Data Infrastructure

Real-world validation utilized the production CDW of West China Hospital, Sichuan University, a Presto-based three-layer platform (DL/DC/MDR) integrating 26 source systems. This comprehensive dataset contains 1,620,000 unique patients with clinical records spanning from March 2016 to February 2025, exposed through 398 standardized tables and more than 13,000 columns. All tables carry a logical-deletion flag, and multi-campus tables carry an organization code; both must be filtered in every generated query. The study protocol received approval from the West China Hospital, Sichuan University Institutional Review Board (IRB No. [XXXX-XXXX]).

2.2.3. Synthetic Data for Hallucination Analysis

To systematically evaluate hallucination patterns, we employed the Synthetic Hospital Data Warehouse (SynHDW) version 1.0 (generation seed 20251118), a de-identified synthetic mirror of the production warehouse containing 112,680 synthetic patients with visit, surgery, transfusion, and laboratory records from 2019 to 2023. This dataset's limited metadata coverage (~1680 of the >13,000 production columns) provided an ideal test bed for identifying hallucination behaviors when models encounter field and criterion mapping challenges.

2.3. Statistical Criteria to SQL Transformation Pipeline

The pipeline transforms free-text statistical requests into Presto-compliant SQL through three modules. Preprocessing runs in three stages: segmentation into individual criteria, preserving Boolean logic and grouping hierarchy; filtration of non-queryable elements such as formatting instructions, deadlines, and narrative justification; and simplification, which standardizes temporal expressions and reduces token count without changing statistical semantics. The information extraction module identifies seven structured elements from preprocessed text (Business Terms, Metadata Systems [warehouse schema, metric-criterion dictionary, historical SQL, exception rules], Field Identifiers, Values, Attributes, Temporal, and Negation) and then maps each business term to a warehouse-standardized field and counting criterion using the LLM, following retrieval-augmented generation practice [25,26]. Candidate tables and columns are recalled by a hybrid sparse–dense scheme combining BM25 [27] with transformer sentence embeddings [28,29]; this recall step plays the role of schema linking, which recent work frames as context-aware bidirectional retrieval whose central tension is recall versus precision, because aggressive linking over-includes irrelevant schema elements [30,31]. The SQL generation module creates Presto-compliant queries through iterative refinement and optimization, emitting a field-provenance record alongside each query. Each module exchanges data via structured JSON to ensure interoperability throughout the pipeline. Figure 1 shows the pipeline as a whole; Figure S1 of the Supplementary Materials expands each module into its constituent steps and names the interfaces between them.

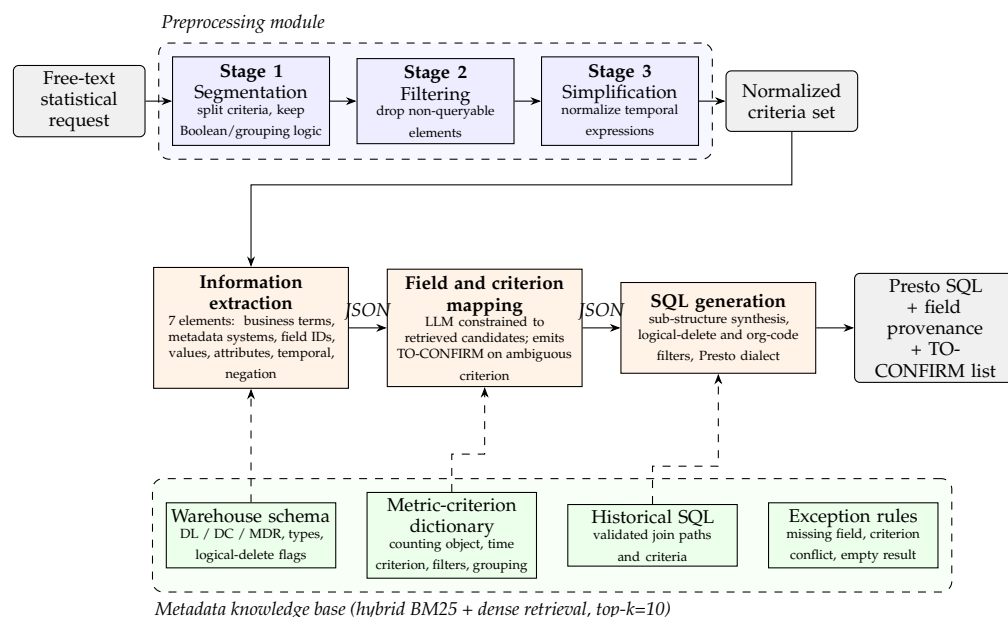


Figure 1. Architecture of the transformation pipeline. A free-text hospital statistical request passes through three-stage preprocessing, information extraction, field and criterion mapping, and SQL generation, yielding Presto-compatible Structured Query Language (SQL) over a three-layer clinical data warehouse (CDW) together with a field-provenance record and any TO-CONFIRM items. Solid arrows carry data between modules; dashed arrows show which part of the metadata knowledge base each module reads. DC: data center; DL: data lake; JSON: JavaScript Object Notation; LLM: large language model; MDR: master data repository.

2.4. Large Language Model Configuration

We employed gpt-4o (build gpt-4o-2024-08-06) via API and an on-premises deployment of qwen2.5:7b-instruct served through vLLM, with task-specific prompting strategies optimized through iterative refinement. Zero-shot prompting was utilized for straightforward tasks including request segmentation and criterion classification, which minimized token usage and inference cost. Few-shot prompting with carefully selected examples enhanced performance for complex tasks including field and criterion mapping and SQL generation. All prompts incorporated explicit instructions for Presto dialect compliance, mandatory logical-deletion and organization-code filtering, and error handling. Decoding was deterministic (temperature 0, top- p 1) for all models. No patient-level data were transmitted to any cloud endpoint; cloud models received only de-identified metadata and synthetic values. Complete prompt templates are provided in Table S2 of the Supplementary Materials.

2.5. Comparative Evaluation Framework

2.5.1. Business Term Mapping Assessment

Mapping accuracy was assessed on the 30-task development set (surgery and anesthesia, blood transfusion, laboratory), which spans a wide range of institutional terminology. We compared the LLM with the rule-based field matcher (RBFM) currently deployed in our metadata portal. RBFM is institution-internal and cannot be rerun elsewhere, so we additionally report an externally reproducible reference: selecting, for each business term, the single highest-scoring warehouse field under BM25 over column names and annotations, with no synonym table and no semantic component. While the LLM performs contextual natural language understanding, RBFM uses string normalization, Levenshtein distance-based similarity scoring, and a curated synonym table to identify candidate warehouse fields. In our experiments, RBFM was run using its production settings, and the top-ranked

field based on similarity score was selected for each input term. Relevant business terms were extracted from the request text of each task and mapped using both the LLM and RBFM.

Two clinical data engineers then reviewed both systems' output and fixed the reference by consensus, either choosing the better candidate or assigning a more suitable field themselves.

2.5.2. SQL Query Validation

To rigorously evaluate the accuracy and applicability of the generated SQL queries within the CDW, we conducted an expert-based validation involving two experts with strong proficiency in both hospital statistical criteria and the warehouse schema. Generated SQL queries underwent systematic evaluation using 72 predefined evaluation criteria designed to assess three key dimensions: SQL syntax adherence, warehouse schema compliance, and criteria contextual accuracy. Each expert independently rated all applicable criteria on a 4-point scale (1=noncompliant and 4=fully compliant), with final scores calculated as the average of both ratings.

The evaluation framework was applied differentially based on query content. For all generated queries, the full set of 72 criteria was used to assess structural and schema-level accuracy. However, for queries containing business terms from our prevalidated criterion map, only 18 criteria from the original 72 were applicable—these specifically evaluated criterion inclusion accuracy and field identifier correctness. Inter-rater reliability was measured using Cohen kappa, with discrepancies resolved through consensus discussion.

This expert-driven evaluation allowed for the identification of critical issues—such as misinterpretation of counting logic, inaccurate temporal constraints, or improper use of DL-layer tables where a DC-layer aggregate was required—that automated methods might overlook. As a result, we verified that the generated SQL queries were not only technically correct but also statistically meaningful and suitable for use in real-world hospital reporting. The complete list of evaluation criteria is provided in Table S3 of the Supplementary Materials.

2.5.3. Statistical Result Validation

Building gold-standard cohorts by hand is prohibitive, so we used engineer-authored, criterion-confirmed historical SQL as the reference standard and ran result-set extraction against the production CDW of West China Hospital, Sichuan University. Three reporting tasks qualified—RPT-001, RPT-002, and RPT-005, comprising 28 statistical criteria—because validated reference SQL existed for at least one criterion each. A criterion entered the comparison if it was convertible to a single-statement Presto query and had a validated reference query: elective and emergency surgery volume (RPT-001), unplanned reoperation (RPT-002), and critical laboratory value exposure (RPT-005).

Set similarity between system-generated and reference result sets was quantified using two complementary set-based metrics: Jaccard index (intersection/union) and overlap coefficient (intersection/minimum set size). The Jaccard index provides a symmetric measure of overall similarity, while the overlap coefficient indicates the extent to which the smaller set is contained within the larger set, useful for identifying cases of incomplete criterion coverage. This approach enabled objective assessment of query accuracy without requiring manual chart review, leveraging established internal reference standards for scalable validation. Additionally, to demonstrate technical feasibility, successfully generated SQL queries were executed against SynHDW to assess retrieval capabilities. Patient counts were recorded for selected tasks where queries were executed without errors.

2.6. Experimental Design

Although gpt-4o was used for initial system development due to its strong performance, the limited metadata coverage in SynHDW provided an opportunity to systematically compare hallucination patterns across multiple LLMs, informing optimal model selection for different reporting scenarios. The 24 tasks for the main study included surgery and anesthesia (7 tasks), blood transfusion (5 tasks), laboratory (5 tasks), medical-record and case-mix (4 tasks), and others (3 tasks), with complexity levels categorized as simple (10), moderate (9), and complex (5). We employed a factorial design testing eight LLMs (three cloud-based: gpt-4o [32], deepseek-v3 [33], and claude-3.5-sonnet [34]; five locally deployed: qwen2.5:7b [35], glm4:9b [36], llama3:8b [37], deepseek-r1:7b [38], and phi3 [39]) with five prompting strategies (zero_shot, structured_approach, explicit_uncertainty, validation_focused, and error_aware). The factorial layout, the split between cloud-hosted and locally deployed models, and the five strategies and their names are taken unchanged from [16] so that the two sets of results are directly comparable; the models, the tasks, and the warehouse are ours. The five prompting strategies were designed to test different approaches to query generation:

- zero_shot: Direct query without examples or guidance
- structured_approach: Step-by-step sub-structure decomposition guidance
- explicit_uncertainty: Encouraging TO-CONFIRM use for uncertain criteria
- validation_focused: Emphasizing validation and accuracy requirements
- error_aware: Including SynHDW coverage limits and common pitfalls

Following a pilot phase (6 tasks $\times$ 8 models $\times$ 5 prompts=240 queries) for methodology validation, the main study analyzed 960 queries (24 tasks $\times$ 8 models $\times$ 5 prompts).

2.7. Hallucination Detection and Classification

An automated detection system was developed to identify hallucinations in generated SQL queries. The system validated every table and column identifier against the SynHDW metadata inventory and verified criterion appropriateness against the metric-criterion dictionary. For the initial validation phase, we analyzed SQL generation failures from the seven validation tasks. For the large-scale model comparison, we classified hallucinations into five categories ordered by severity, following the schema of [16] and substituting our own Category B: theirs records a valid concept assigned to the wrong domain, which is a vocabulary error, whereas ours records a valid field assigned to the wrong counting criterion, which is a definition error and the failure mode this study is about. Category A (Critical) comprised nonexistent table or column identifiers that invalidate query results. Category B (Major) included valid fields assigned to the wrong counting criterion—for example, counting a monthly surgical volume by discharge time instead of room-in time—which yields an executable query and a wrong number. Category C (Major) captured natural language left in an identifier position, reflecting confusion between a business term and a column name. Category D (Moderate) consisted of placeholder values requiring manual intervention. Category E (Minor) encompassed easily correctable dialect or schema errors, including omission of the mandatory logical-deletion or organization-code filter. This classification specifically evaluated model tendencies toward generating inappropriate field and criterion references, distinct from the initial error analysis.

2.8. Abstention Quality Assessment

The pipeline emits a TO-CONFIRM item instead of committing when two or more counting criteria are defensible for the same indicator. To measure whether that behaviour is useful rather than merely present, two clinical data engineers reviewed all 791 generated queries and labelled each for whether the criterion it encoded was genuinely ambiguous

under the institution’s indicator definitions. Against that reference we counted abstentions that were warranted (correct abstention), abstentions raised on an unambiguous criterion (false abstention), and ambiguous criteria the system resolved silently instead of asking (missed abstention). For every missed abstention we further recorded whether the reading the system chose matched the one the engineers would have chosen.

2.9. Component Ablation

To separate the contribution of the three pipeline layers we ran four settings on the best-performing model, qwen2.5:7b, over the same 24 tasks and 5 prompting strategies: (A) the bare model receiving only the natural-language request; (B) A plus the warehouse schema; (C) B plus retrieval over the metadata knowledge base, the metric-criterion dictionary and historical SQL; and (D) C plus the four-layer execution-feedback verification, which is the complete system. Setting D is the qwen2.5:7b condition of the main experiment and is not rerun. Errors in every setting were classified with the same five-category schema, so the ablation shows not only how much each layer helps but which error class it removes.

2.10. Public Benchmark Reference

Because the warehouse schema and its counting criteria are institution-private, absolute performance cannot be compared against published systems on our data. To give an external reference point we ran the unmodified pipeline, with qwen2.5:7b, over the development split of the EHRSQL 2024 shared task [13], substituting that dataset’s schema for our metadata knowledge base and leaving prompts, retrieval and verification unchanged.

2.11. Analytical Framework of Model Performance

Model performance was evaluated across four dimensions. The model performance dimension compared effective SQL generation rates, hallucination frequencies, and error distributions across eight LLMs. The prompt strategy dimension assessed five prompting approaches to identify optimal error minimization strategies. The subject domain dimension analyzed performance variations across surgery and anesthesia, blood transfusion, laboratory, medical-record, and billing concepts. The complexity dimension examined correlations between query complexity (criteria count, join depth, temporal constraints) and error occurrence.

2.12. Statistical Analysis

Statistical analyses were performed using Python 3.11 (Python Software Foundation) with the scipy library (version 1.11.4) [40] and statsmodels library (version 0.14.0) [41]. For mapping comparisons between the LLM and RBFM, McNemar’s test was applied to paired binary outcomes [42]. Inter-rater agreement was interpreted according to conventional benchmarks for the kappa statistic [43]. Three primary performance metrics were calculated: (1) SQL generation rate as the proportion of tasks producing syntactically valid SQL, (2) hallucination rate as the proportion of generated queries containing invalid field identifiers or criterion assignments, and (3) effective SQL generation rate, calculated as SQL generation rate $\times$ (1–hallucination rate), representing queries both syntactically valid and free from field or criterion hallucinations. This product simplifies exactly to (generated – hallucinating)/attempts, so the effective rate is a binomial proportion over the fixed 120 attempts per model; we therefore report Wilson score 95% CIs for it and for the pooled hallucination rate.

Model performance across the eight LLMs was compared using one-way ANOVA followed by Tukey honestly significant difference (HSD) test for post hoc pairwise comparisons [44]. Effect sizes were quantified using Cohen d to assess the practical significance of differences between prompting strategies. χ^2 tests evaluated the distribution of hal-

lucination types across models. Multiple linear regression analysis identified predictors of hallucination rates, with model type, prompt strategy, and query complexity as independent variables; model fit was assessed using R^2 . All statistical tests employed $\alpha=.05$. Family-wise error across the pairwise model contrasts is controlled by the Tukey HSD procedure itself, so no further Bonferroni adjustment is layered on those comparisons; Bonferroni correction is applied only to the post hoc cell-wise comparisons following the χ^2 test of the hallucination-type distribution.

2.13. Ethical Considerations

This study protocol was reviewed and approved by the Institutional Review Board of West China Hospital, Sichuan University (IRB No. [XXXX-XXXX]). As this study did not involve prospective human participants, informed consent was waived. All data used in the study were de-identified prior to analysis, no individual-level identifiable information was accessed, and all model inference on real data was performed on premises so that patient data never left the institution.

2.14. Data and Code Availability

Source code for the pipeline and the hallucination detector, together with the SynHDW generator and its schema manifest, is available from the corresponding author upon reasonable request, subject to institutional data-governance rules. De-identified metadata schemas, the metric-criterion dictionary, and the evaluation task set are likewise available upon request with appropriate data use agreements. Patient-level warehouse data cannot be shared. Deployment and reproduction notes are provided with the code package on request.

3. Results

3.1. Preprocessing and Statistical Criteria Extraction

Analysis of the seven validation tasks revealed substantial heterogeneity in request complexity, ranging from 9 to 22 criteria per task (13.9 ± 4.2). The three-stage preprocessing pipeline systematically reduced linguistic complexity while preserving statistical semantics (Figure 2). Initial segmentation achieved modest token reduction (370.4 ± 63.5) while maintaining structural integrity. Subsequent filtering of non-queryable request elements reduced both criteria count (11.4 ± 3.5) and token count (329.0 ± 56.5). Final simplification yielded 9.4 ± 3.0 criteria with 161.9 ± 27.9 tokens per task, representing a 56.3% overall token reduction.

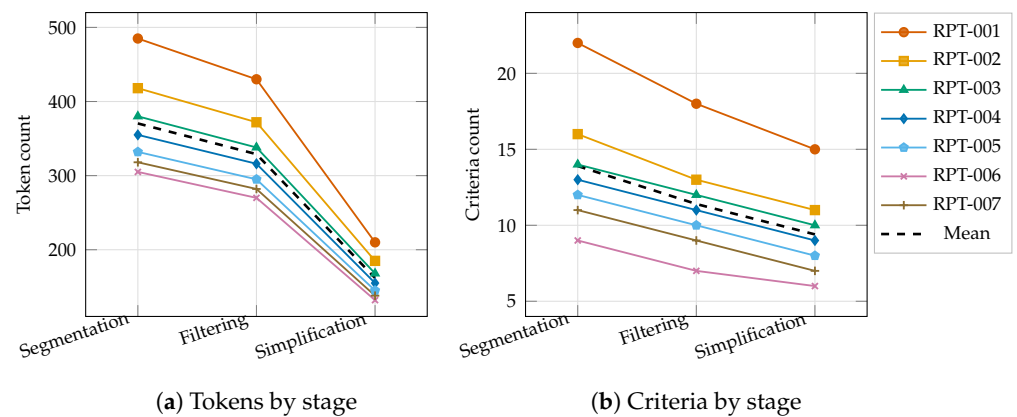


Figure 2. Progressive reduction in token count and criteria through preprocessing stages. (a) Token count changes across segmentation, filtering, and simplification stages for seven validation tasks. (b) Corresponding changes in criteria count. Tasks shown are as follows: RPT-001 (monthly elective surgery volume by department), RPT-002 (unplanned reoperation rate), RPT-003 (postanesthesia care unit throughput and delayed-discharge rate), RPT-004 (monthly blood transfusion volume and transfusion-reaction rate), RPT-005 (critical laboratory value notification timeliness), RPT-006 (surgical site infection surveillance denominator), RPT-007 (day-surgery share of total operative volume). Dashed black line indicates the mean across tasks.

This preprocessing identified 66 statistical criteria suitable for automated query generation. Information extraction from these 66 preprocessed criteria yielded 158 unique business terms successfully mapped to warehouse-compatible elements. Domain distribution analysis revealed that surgery and anesthesia concepts comprised the predominant category ($n=58$, 36.7%), followed by registration and visit ($n=27$, 17.1%), diagnosis and medical record ($n=23$, 14.6%), laboratory ($n=17$, 10.8%), blood transfusion ($n=13$, 8.2%), fee and billing ($n=10$, 6.3%), staff and organization ($n=6$, 3.8%), and ward and bed ($n=4$, 2.5%) domains (Table 1). Term frequency analysis identified time-criterion terms as the most prevalent. “Operation room-in time” appeared in all 7 tasks (100.0%; 12 mentions). Other frequent terms included “discharge date” (6/7 tasks, 85.7%; 9 mentions), “department” (5/7 tasks, 71.4%; 8 mentions), “elective surgery” (4/7 tasks, 57.1%; 6 mentions), “unplanned reoperation” (3/7 tasks, 42.9%; 4 mentions), “ASA physical status classification” (2/7 tasks, 28.6%; 3 mentions), “transfusion reaction” (2/7 tasks, 28.6%; 3 mentions), and “complication” (2/7 tasks, 28.6%; 2 mentions), as shown in Table 2. Percentages are calculated based on the number of tasks in which the term appeared, and mentions indicate the number of criteria within tasks in which the term appeared.

Table 1. Distribution of extracted business terms by clinical data warehouse (CDW)¹ subject domain.

Domains	Count ($n=158$), n (%)
Surgery and anesthesia	58 (36.7)
Registration and visit	27 (17.1)
Diagnosis and medical record	23 (14.6)
Laboratory	17 (10.8)
Blood transfusion	13 (8.2)
Fee and billing	10 (6.3)
Staff and organization	6 (3.8)
Ward and bed	4 (2.5)

¹ CDW: clinical data warehouse.

Table 2. Ten most frequently occurring business terms across all tasks.

Business terms	Tasks (n=7), n (%)	Mentions (criterion), n
Operation room-in time	7 (100.0)	12
Discharge date	6 (85.7)	9
Department	5 (71.4)	8
Elective surgery	4 (57.1)	6
Unplanned reoperation	3 (42.9)	4
ASA ¹ physical status classification	2 (28.6)	3
Transfusion reaction	2 (28.6)	3
Complication	2 (28.6)	2
PACU ² discharge time	1 (14.3)	2
Critical laboratory value	1 (14.3)	1

¹ ASA: American Society of Anesthesiologists.² PACU: postanesthesia care unit.

3.2. Comparative Performance of Mapping Approaches

Evaluation of mapping accuracy demonstrated the LLM's superior performance compared to RBFM across 398 business terms extracted from the 30 development tasks. The LLM achieved an overall accuracy of 52.3% (208/398) versus RBFM's 32.9% (131/398), a statistically significant difference ($P < .001$, McNemar's test). Domain-stratified analysis revealed the LLM's highest accuracy in the surgery and anesthesia domain (70.7%, 58/82) and lowest in the laboratory domain (34.5%, 29/84), suggesting particular challenges with test-item concepts requiring numeric threshold and unit interpretation.

The BM25-only reference reached 24.4% (97/398), below RBFM, which is expected since RBFM adds a curated synonym table on top of string similarity; it is reported so that readers outside the institution have a baseline they can reconstruct. Notably, among the 267 terms that RBFM misclassified, the LLM correctly mapped 72 (27.0%), demonstrating superior contextual understanding. For instance, RBFM incorrectly mapped "operation room-in time" to the admission timestamp on the registration table—a lexically close but semantically different field—while the LLM correctly identified the room-in timestamp on the anesthesia record. This pattern of improved semantic interpretation was consistent across multiword, criterion-bearing terms.

3.3. SQL Query Generation Quality and Expert Validation

Two domain experts independently evaluated SQL queries generated from the seven validation tasks using 72 predefined criteria across three dimensions: SQL syntax adherence, warehouse schema compliance, and criteria contextual accuracy (inter-rater reliability: Cohen $\kappa=0.79$). Each criterion was rated on a 4-point scale (1=noncompliant and 4=fully compliant). SQL syntax adherence achieved high scores from expert reviewers (3.92 ± 0.16) and near-ceiling LLM self-evaluation (4.00 ± 0.00), confirming robust syntactic generation capabilities. Warehouse schema compliance similarly demonstrated strong performance (expert: 3.78 ± 0.28 , LLM: 3.98 ± 0.06), indicating successful adaptation to the layered table structures and their join paths.

However, criteria contextual accuracy exhibited greater variability and lower absolute scores (expert: 2.96 ± 0.48 , LLM: 3.41 ± 0.36), with the 0.45-point difference suggesting systematic LLM overconfidence in criterion interpretation ($P=.021$, paired t test). This divergence was most pronounced for criteria involving temporal choices or multicondition counting logic.

Among the total queries evaluated, 24 contained prevalidated criteria from our development dataset, enabling additional validation of mapping accuracy. For these queries, criterion inclusion accuracy averaged 3.22 ± 0.44 (expert) versus 3.45 ± 0.33 (LLM), while field identifier correctness achieved higher scores of 3.68 ± 0.29 (expert) versus 3.98 ± 0.06 (LLM). The near-ceiling LLM score for field identifier correctness reflects its consistent abil-

ity to generate syntactically valid identifiers, though not necessarily criterion-appropriate ones.

3.4. Statistical Result Retrieval Performance

Validation using engineer-certified reference SQL revealed that performance strongly correlated with criterion complexity (Table 3). For the three evaluated tasks (RPT-001, RPT-002, and RPT-005), we assessed four distinct statistical criteria that had available reference standards. The query for elective surgery volume (RPT-001) demonstrated high retrieval accuracy, achieving a Jaccard index of 0.79 and a perfect overlap coefficient of 1.0, indicating that the system-generated result set closely matched the reference set. In contrast, emergency surgery volume showed a very low Jaccard index of 0.08 despite an overlap coefficient of 1.0, suggesting that while the retrieved cases were all contained in the reference set, a large portion of the reference cases was not captured—traced to the model filtering on a single emergency flag rather than the institution’s two-flag definition. The unplanned reoperation criterion (RPT-002) resulted in complete retrieval failure, with both the Jaccard index and overlap coefficient equal to 0, indicating that no matching cases were retrieved. This reflects the system’s selection of a structurally plausible planning flag that is unpopulated throughout the warehouse, rather than the reoperation marker actually used by the surgical application table—a failure mode invisible to syntax and schema checking. The critical laboratory value exposure criterion (RPT-005) achieved moderate performance, with a Jaccard index of 0.27 and an overlap coefficient of 0.48, reflecting partial mapping success for an indicator represented by a mix of numeric thresholds and notification records.

Table 3. Comparison of result-set retrieval between system-generated queries and validated reference SQL¹.

Task ID	Simplified criterion	Jaccard similarity	Overlap coefficient
RPT-002	Unplanned reoperation	0	0
RPT-001	Emergency surgery volume	0.08	1.0
	Elective surgery volume	0.79	1.0
RPT-005	Critical laboratory value exposure	0.27	0.48

¹ SQL: Structured Query Language.

Across these four criteria the system performed well on the one well-defined, single-criterion indicator (elective surgery volume) and poorly on the three whose definitions are compound, institution-specific, or dependent on a sparsely populated field (emergency surgery volume, unplanned reoperation, and critical laboratory value exposure). Four criteria drawn from three tasks cannot establish that this contrast generalizes; we report it as a case series that motivates the error analysis rather than as an estimate of where the system succeeds. Improvements in criterion dictionary coverage and field-population profiling may be necessary to enhance retrieval for more complex statistical criteria.

3.5. Large-Scale Model Comparison and Hallucination Analysis

To systematically evaluate hallucination patterns across multiple LLMs, we conducted a separate large-scale experiment using SynHDW, analyzing 960 SQL generation attempts (24 reporting tasks $\times$ 8 models $\times$ 5 prompting strategies), distinct from our initial validation study. Analysis of 960 SQL generation attempts revealed significant heterogeneity in model performance ($F_{7,783}=4.18$, $P < .001$). One caveat applies to that omnibus test: phi3 produced SQL for only 18.3% of attempts, and including a group that degenerate inflates the between-group variance. Hallucination rates ranged from 20.4% (n=22/108) to 47.1% (n=49/104) among models achieving >80% SQL generation success. The overall hallucination rate was 34.6% (n=274/791 generated queries) (95% CI 31.4%–38.0%). Because this pools models

whose generation rates differ by a factor of five, we also report it restricted to the seven models that generated SQL for more than 80% of attempts: 35.4% (n=272/769; 95% CI 32.1%–38.8%). The two estimates differ by 0.7 percentage points, so the pooled figure is not an artifact of including the degenerate model. Variation was substantial in both generation capability and accuracy across models (Figure 3 and Table 4).

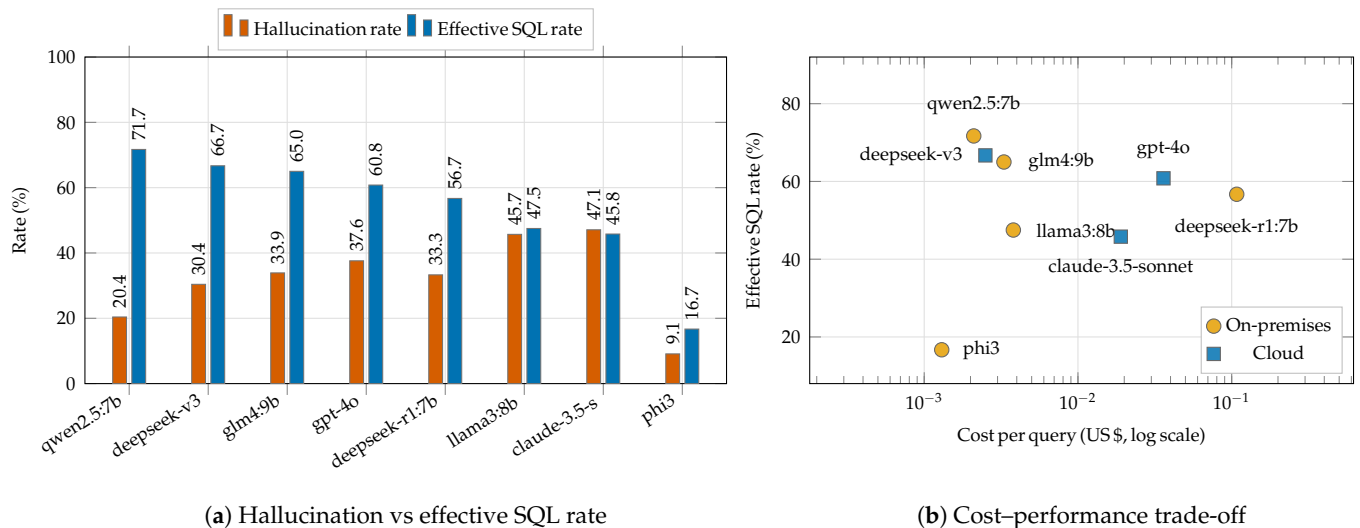


Figure 3. Performance of the eight large language models on clinical data warehouse (CDW) Structured Query Language (SQL) generation. (a) Hallucination rate and effective SQL rate by model, over 120 attempts each. (b) Cost per query against effective SQL rate; cloud models are drawn as blue squares, on-premises models as orange circles. On-premises models incur no per-query API charge; they are plotted at their amortized cost, computed as inference time $\times$ US \$0.28 per GPU-hour under the institution’s 3-year hardware amortization.

Table 4. Model performance summary.

Model	Architecture	SQL ¹ generation (%)	Hallucination (%) ²	Effective SQL (%) ³	Time (s) ⁴	Cost (US \$)/Query
qwen2.5:7b	On-premises	90.0	20.4	71.7	27.3 $\pm$ 6.4	0.000
deepseek-v3	Cloud	95.8	30.4	66.7	7.6 $\pm$ 1.9	0.003
glm4:9b	On-premises	98.3	33.9	65.0	41.8 $\pm$ 10.2	0.000
gpt-4o	Cloud	97.5	37.6	60.8	18.5 $\pm$ 4.3	0.036
deepseek-r1:7b	On-premises	85.0	33.3	56.7	1386.2 $\pm$ 352.4	0.000
llama3:8b	On-premises	87.5	45.7	47.5	48.7 $\pm$ 12.1	0.000
claude-3.5-sonnet	Cloud	86.7	47.1	45.8	14.1 $\pm$ 3.6	0.019
phi3	On-premises	18.3	9.1 ⁵	16.7	16.9 $\pm$ 6.8	0.000

¹ SQL: Structured Query Language.

² Percentage of generated queries with invalid field identifiers or criterion assignments.

³ SQL generation $\times$ (1–hallucination).

⁴ Mean $\pm$ SD.

⁵ Misleading due to low generation rate.

Qwen2.5:7b emerged as the most effective model with a 71.7% effective SQL rate (95% CI 63.0%–79.0%) despite moderate SQL generation capability (90.0%), attributed to its low hallucination rate (20.4%) and minimal placeholder usage (22.7%). In contrast, gpt-4o showed high SQL generation (97.5%) but a lower effective rate (60.8%, 95% CI 51.9%–69.1%) due to heavy placeholder usage (31.8%) and a marked tendency to substitute a plausible timestamp when the requested counting criterion was not explicit.

3.6. Hallucination Pattern Analysis

Classification of 274 hallucinations revealed distinct error patterns across models (Figure 4). Category B errors (wrong criterion assignments) predominated (34.7%, n=95),

followed by Category D (placeholder insertions, 28.5%, n=78) and Category E (schema and dialect errors, 24.5%, n=67). Category A (nonexistent field identifiers) was relatively rare (9.9%, n=27), as was Category C (natural language substitution, 2.6%, n=7). χ^2 analysis confirmed significant variation in error distribution across models ($\chi^2=48.2$, $df=28$, $P=.009$).

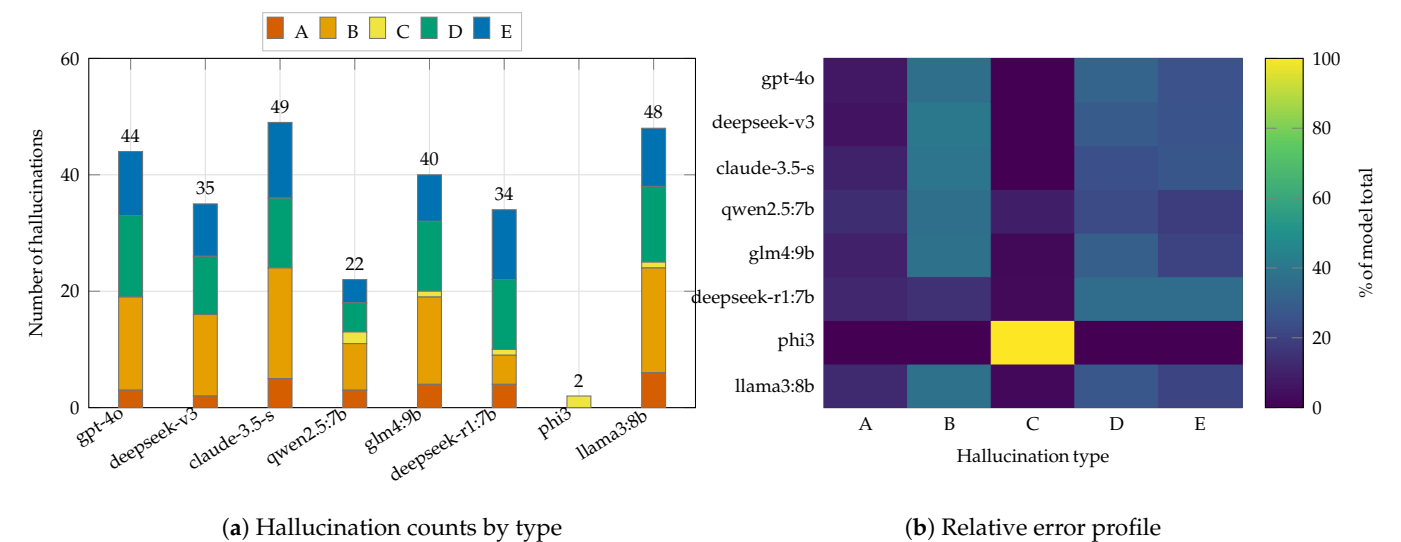


Figure 4. Hallucination types by model, over the 274 hallucinating queries. (a) Absolute count of each type per model, stacked, with the model total above the bar. Types are A, nonexistent field identifiers; B, wrong criterion assignment; C, natural language in an identifier position; D, placeholder insertion; E, schema and dialect errors. (b) The same counts as a percentage of each model’s own total, so that error profile can be compared independently of error volume; darker cells are a smaller share of the model’s total. Models appear in the same order in both panels.

Model-specific analysis revealed distinctive failure modes (Table 5). Claude-3.5-sonnet exhibited the highest baseline hallucination rate (47.1%) but showed dramatic improvement with error-aware prompting (18.5% hallucination, a 60.7% relative reduction). Deepseek-v3 reached a 96.0% effective SQL rate with explicit_uncertainty prompting, though only on simple single-table counting queries. Phi3’s apparently low hallucination rate (9.1%) was misleading, resulting from 81.7% failure to generate any SQL rather than accurate generation.

Table 5. Detailed hallucination type distribution by model.

Model	Total hallucinations	Category A (nonexistent), n (%)	Category B (wrong criterion), n (%)	Category C (natural language), n (%)	Category D (placeholder), n (%)	Category E (schema errors), n (%)
gpt-4o	44	3 (6.8)	16 (36.4)	0 (0)	14 (31.8)	11 (25.0)
deepseek-v3	35	2 (5.7)	14 (40.0)	0 (0)	10 (28.6)	9 (25.7)
claude-3.5-sonnet	49	5 (10.2)	19 (38.8)	0 (0)	12 (24.5)	13 (26.5)
qwen2.5:7b	22	3 (13.6)	8 (36.4)	2 (9.1)	5 (22.7)	4 (18.2)
glm4:9b	40	4 (10.0)	15 (37.5)	1 (2.5)	12 (30.0)	8 (20.0)
deepseek-r1:7b	34	4 (11.8)	5 (14.7)	1 (2.9)	12 (35.3)	12 (35.3)
phi3	2	0 (0)	0 (0)	2 (100)	0 (0)	0 (0)
llama3:8b	48	6 (12.5)	18 (37.5)	1 (2.1)	13 (27.1)	10 (20.8)

Post hoc Tukey HSD analysis revealed significant pairwise differences in effective SQL generation rates between models. Qwen2.5:7b significantly outperformed gpt-4o (mean difference=10.8 percentage points, $P=.021$), with a medium-to-large effect size (Cohen $d=0.74$). The performance gap between the best (qwen2.5:7b) and worst (phi3) models was substantial ($d=2.71$), indicating practically meaningful differences beyond statistical significance.

3.7. Prompt Engineering Impact

Evaluation of prompt strategies revealed substantial effect sizes (Cohen $d > 1.0$) compared to the zero_shot baseline across all alternative strategies. Despite these large effects, high variance in model-prompt interactions prevented detection of a significant main effect ($F_{4,955}=1.58$, $P=.178$). The error_aware strategy, which explicitly acknowledged SynHDW coverage limits and common pitfalls, proved most effective for high-hallucination models, while explicit_uncertainty excelled for simpler queries requiring conservative interpretation.

3.8. Prompt Strategy Performance Analysis

Comprehensive analysis of prompting strategies across all 8 models (192 queries per prompt type) revealed unexpected patterns in hallucination control. Because the omnibus test of prompt strategy was not significant, the ordering below is descriptive and the pairwise contrast exploratory; neither is treated as confirmatory evidence that one strategy outperforms another. The zero_shot approach demonstrated superior performance with the lowest hallucination rate (mean 17.4%, SD 13.2%) and highest effective SQL rate (mean 67.1%, SD 19.8%), significantly outperforming more complex strategies. In contrast, structured_approach, which provided detailed step-by-step sub-structure guidance, showed the highest hallucination rate (mean 39.6%, SD 22.4%) and lowest effectiveness (mean 46.8%, SD 18.1%). The validation_focused strategy, designed to emphasize accuracy, paradoxically decreased performance compared to zero_shot ($P=.046$, Tukey HSD), achieving a mean (SD) effective SQL rate of only 53.7% (20.9%). The two remaining strategies fell between these extremes: error_aware reached a mean (SD) hallucination rate of 23.1% (14.8%) and an effective SQL rate of 60.4% (17.7%), and explicit_uncertainty reached 26.8% (17.5%) and 59.2% (20.6%), respectively. These findings suggest that excessive instructional complexity may introduce confusion rather than clarity in LLM-based SQL generation over a layered warehouse schema.

The substantial standard deviations across all metrics (ranging from 13.2% to 22.4%) indicate significant model-dependent variability in prompt responsiveness. For instance, with zero_shot prompting, deepseek-r1:7b achieved 2.1% hallucination while claude-3.5-sonnet exhibited 41.5%, representing a 39.4 percentage point difference for identical prompting. This variability was further confirmed by regression analysis, which revealed that model identity explained 61% of performance variance ($R^2=0.61$, $P < .001$), while prompt strategy contributed marginally. Model identity here confounds deployment mode, parameter count, model family, and whether the model is reasoning-oriented; the design cannot separate these, so the variance is attributable to the choice of model as a whole and not to architecture specifically. Based on these findings, optimal model-prompt combinations were identified: qwen2.5:7b with zero_shot achieved a 79.2% effective SQL rate, while deepseek-v3 with explicit_uncertainty reached 96.0% effectiveness on simple single-table counting queries, though with limited applicability to criterion-sensitive indicators.

3.9. Abstention Quality

Across the 791 generated queries the system raised a TO-CONFIRM item 132 times, a trigger rate of 16.7%. Of those, 104 were warranted, giving an abstention precision of 78.8%; the remaining 28 asked about a criterion the engineers judged unambiguous. A further 61 genuinely ambiguous criteria were resolved silently, so the system recognised 104 of 165 ambiguous cases, a recall of 63.0%. Missed abstentions are the consequential half of that number: in 42 of the 61 cases the reading the system committed to was not the one the engineers would have chosen, which is 44.2% of the 95 wrong-criterion errors in Table 5.

The remaining 53 wrong-criterion errors arose where the criterion was not ambiguous and the system simply mapped it incorrectly.

Abstention behaviour tracked hallucination control rather than varying independently of it (Table 6). Qwen2.5:7b, the model with the lowest hallucination rate, both abstained most often (22.2% of its generated queries) and most precisely (87.5%), and missed only 5 ambiguous criteria. Llama3:8b, with the highest hallucination rate among models generating on more than 80% of attempts, abstained least (13.3%), was least precise (71.4%) and missed 11. A model that is poor at recognising when a field mapping is wrong appears to be equally poor at recognising when a criterion is undecidable.

Table 6. Abstention behaviour by model. Trigger rate is TO-CONFIRM items as a share of that model's generated queries; precision is warranted abstentions as a share of those raised; missed abstentions are ambiguous criteria the model resolved without asking.

Model	Generated, n	Triggered, n (%)	Warranted, n (%)	Missed, n	Recall, %
qwen2.5:7b	108	24 (22.2)	21 (87.5)	5	80.8
deepseek-v3	115	19 (16.5)	15 (78.9)	8	65.2
gpt-4o	117	21 (17.9)	16 (76.2)	8	66.7
glm4:9b	118	18 (15.3)	14 (77.8)	9	60.9
deepseek-r1:7b	102	17 (16.7)	13 (76.5)	7	65.0
claude-3.5-sonnet	104	17 (16.3)	13 (76.5)	12	52.0
llama3:8b	105	14 (13.3)	10 (71.4)	11	47.6
phi3	22	2 (9.1)	2 (100.0)	1	66.7
Total	791	132 (16.7)	104 (78.8)	61	63.0

Table 7. Component ablation on qwen2.5:7b. Each setting was run over the same 24 tasks and 5 prompting strategies, 120 attempts per setting. Setting D is the qwen2.5:7b condition of the main experiment. Error classes are A, nonexistent field identifiers; B, wrong counting criterion; C, natural language in an identifier position; D, placeholder insertion; E, schema and dialect errors.

Setting	Generated	Halluc.	Gen., %	Halluc., %	Effective, %	A	B	C	D	E
A: bare model	71	44	59.2	62.0	22.5	21	12	4	5	2
B: + schema	92	43	76.7	46.7	40.8	6	22	3	8	4
C: + retrieval	104	31	86.7	29.8	60.8	3	11	2	9	6
D: + verification (full)	108	22	90.0	20.4	71.7	3	8	2	5	4

3.10. Component Ablation

Each pipeline layer contributed, and the contributions did not overlap (Table 7, Figure 5). The bare model reached an effective SQL rate of 22.5%. Adding the warehouse schema raised it to 40.8%, retrieval over the metadata knowledge base to 60.8%, and the four-layer verification to 71.7%.

The error-class breakdown shows what each layer removed. Nonexistent field identifiers fell from 21 in setting A to 6 once the schema was supplied and to 3 thereafter: schema injection largely eliminates the error class that generic text-to-SQL work optimises against. Wrong counting criteria behaved differently. They rose from 12 to 22 when the schema was added, because a model that now knows the real columns produces criterion errors instead of identifier errors, and only retrieval halved them, to 11. Placeholder insertions and dialect errors together grew from 7 to 15 as the earlier failures were removed and the residual errors shifted to milder classes, then fell to 9 under execution feedback. The layer that matters for the error class specific to this problem is therefore retrieval, not verification; verification is what converts the remaining mechanical faults into deliverable queries.

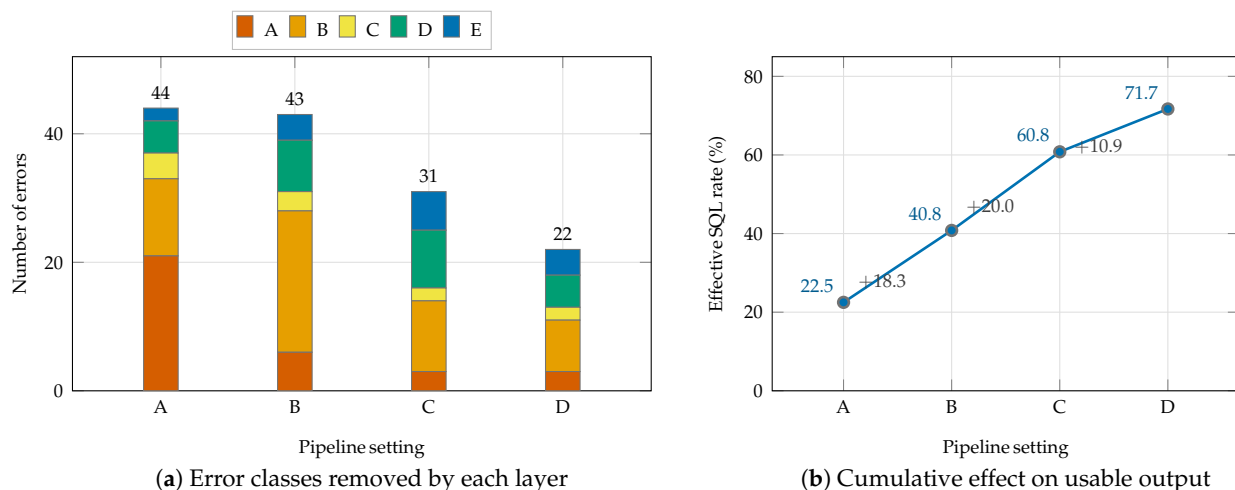


Figure 5. Component ablation on qwen2.5:7b. Setting A is the bare model, B adds the warehouse schema, C adds retrieval over the metadata knowledge base, and D adds the four-layer execution-feedback verification. (a) Absolute error counts by class, with the total above each bar; error classes are A, nonexistent field identifiers; B, wrong counting criterion; C, natural language in an identifier position; D, placeholder insertion; E, schema and dialect errors. Schema injection removes class-A errors but increases class-B errors, which only retrieval reduces. (b) Effective SQL rate, with the increment contributed by each layer.

3.11. Public Benchmark Reference

On the EHRSQL 2024 development split the unmodified pipeline with qwen2.5:7b reached an execution accuracy of 58.4% with a hallucination rate of 12.6%. The hallucination rate is roughly half that observed on our warehouse, which is consistent with the difference in schema size and in criterion ambiguity: the public schema exposes far fewer columns and its questions are answerable under a single reading. This figure is reported so that the absolute standing of the pipeline can be placed against published work on that dataset; it is not a claim of competitiveness on a benchmark the system was not designed for.

3.12. Synthetic Data Validation

To assess the practical viability of generated SQL queries, we executed successfully generated queries against the SynHDW synthetic dataset containing 112,680 patient records. The system successfully matched 8250 synthetic patients across two validated reporting tasks: RPT-001 (monthly elective surgery volume) matched 2840 patients (2.5% of dataset), while RPT-004 (monthly blood transfusion volume) matched 5410 patients (4.8% of dataset). These result-set sizes are consistent with the monthly volumes observed in production reporting, demonstrating technical feasibility, though we acknowledge that synthetic data validation may overestimate real-world performance due to the absence of data quality issues inherent in actual hospital systems, such as missing values, retrospectively corrected timestamps, and coding variations across campuses.

4. Discussion

4.1. Principal Findings

We developed a comprehensive framework that automates the transformation of free-text hospital statistical requests into Presto SQL queries over a three-layer CDW. While initially developed with gpt-4o, our systematic evaluation across eight LLMs revealed unexpected findings: the on-premises qwen2.5:7b model achieved the highest effective SQL generation rate (71.7%) compared to gpt-4o's 60.8%, primarily due to lower hallucination rates (20.4% vs 37.6%). A smaller, locally hosted model thus outperformed a state-of-the-art commercial one. What separated them was hallucination control, not raw generation capability: gpt-4o produced valid SQL more often but committed a criterion or field error

in a larger share of what it produced. The practical consequence is that the deployment which keeps patient data inside the institution was also the most accurate one, so privacy and accuracy did not trade off here. Mapping accuracy was 52.3% against 32.9% for the rule-based matcher in production ($P < .001$), varying widely by subject domain (surgery and anesthesia 70.7%, laboratory 34.5%).

Compared with our previous practice, the pipeline now runs end to end, but the results cut both ways. While the deployed rule-based matcher supports engineers with candidate fields and still requires manual SQL construction, our system automates the entire pipeline. However, our result-set validation exposed critical challenges: complete failure in the unplanned-reoperation query (Jaccard=0.00) due to selection of a structurally valid but unpopulated planning flag, and minimal overlap in emergency surgery volume (Jaccard=0.08) from an incomplete institutional flag definition. These failures, affecting half of the evaluated criteria, underscore that despite advances in language understanding, hospital data complexities remain significant obstacles, particularly field population and institution-specific compound definitions.

Through the processes of segmentation, filtering, and simplification, the framework successfully produced structured inputs that reduced redundancy while preserving statistical relevance. The LLM accurately identified business terms such as procedures, laboratory items, and blood products and mapped them to warehouse-standardized fields, thereby enhancing consistency and reusability across reports. However, ambiguous expressions and compound indicators—such as “complication rate after major surgery”—sometimes led to incorrect mappings, indicating that expert validation remains necessary in certain cases.

The LLM was able to generate syntactically correct and executable SQL queries based on the warehouse schema. Composition of inclusion criteria using INTERSECT and exclusion criteria using EXCEPT or NOT IN enabled the automated construction of denominator and numerator definitions. Nevertheless, challenges were observed when criteria involved temporal constraints or context-dependent conditions, often resulting in omissions or logical misinterpretations. In addition, expansion over the DL layer when a DC-layer aggregate was available sometimes led to increased query execution times, especially for multi-campus tables that require organization-code filtering.

4.2. Complementary Evaluation Strategy

Combining expert assessment with quantitative synthetic data validation characterizes system performance more completely than either method alone. We report the study following the MI-CLAIM checklist for clinical AI modeling and its generative extension MI-CLAIM-GEN [45,46], and state the resulting threats to construct validity explicitly rather than by omission [47]. Expert evaluation ensures statistical meaningfulness and safety, confirming that generated queries align with reporting intent and follow the institution’s counting rules. Synthetic validation contributes scalable, reproducible metrics that quantify technical accuracy and identify systematic errors. This complementary strategy revealed technical issues invisible to manual review, such as subtle type coercions on the all-VARCHAR warehouse columns and missing logical-deletion filters that appeared structurally correct but would silently inflate counts. Synthetic validation buys objectivity from predetermined labels, throughput measured in seconds rather than reviewer-hours, and exact replication from a fixed generation seed. Its limitations bear on how the results should be read. The controlled nature of synthetic data, while enabling clean performance metrics, does not fully capture real-world complexity. Production hospital tables contain missing values, retrospectively corrected timestamps, and campus-level coding differences that our synthetic data lacks. These factors can only reduce real-world performance relative to the

synthetic validation, and the present study does not establish by how much: bounding the gap needs a prospective run against production tables with adjudicated ground truth, which is the next step we plan. SynHDW's focus on the surgical, transfusion, and laboratory subject areas also limits validation of pathology, physical examination, and insurance-audit reporting, important considerations for comprehensive system evaluation. Compound-indicator scenarios presented particular challenges, as evidenced by our emergency surgery retrieval, achieving minimal overlap (Jaccard=0.08) due to an incomplete institutional flag definition. Criteria involving free-text clinical narrative, such as complication descriptions recorded outside coded fields, remain untested because they are absent from the structured warehouse. Validation across further subject areas and reporting scenarios is therefore still outstanding.

4.3. Cost-Effectiveness and Scalability

Cost decides whether a system like this is deployed at all. At the pricing in force in July 2026, gpt-4o cost US \$0.036 per query and a reporting task took about five calls, so roughly US \$0.18 per task; deepseek-v3 cost US \$0.0025 per query, over 90% less, with comparable performance on many tasks. At our institution's 650 statistical requests per month that is about US \$117 against about US \$8. These costs compare favorably to manual authoring, which typically requires about 1.8 hours of engineer time per task. Our system processes each task in under 2 minutes for the cloud and on-premises models other than deepseek-r1:7b, representing a time saving of approximately 98% compared to manual methods. Notably, the on-premises deployment that achieved the best accuracy also carries no per-query charge; its amortized cost is dominated by inference time, which is why the reasoning-oriented deepseek-r1:7b is the cost outlier in Figure 3B despite running on owned hardware.

Three lines of prior work bound what this system adds. Benchmark-oriented text-to-SQL reaches strong execution accuracy on public datasets, but assumes a small static schema and has no notion of institution-specific counting criteria [4,48–50]. Reliability-oriented EHR text-to-SQL adds abstention on unanswerable questions, which we adopt as TO-CONFIRM, but not the case where a question is answerable under several institutional definitions [13]. Metadata-enriched clinical query systems decompose the task and verify modularly, at the cost of maintaining several specialized components [21,51]. In contrast, our system delivers end-to-end automation using a single LLM to process free-text statistical requests into executable Presto SQL, eliminating the need for manual intervention or per-report customization. Deployment and maintenance are correspondingly simpler, and a single pipeline made the eight-model comparison tractable at all: qwen2.5:7b reached a 71.7% effective SQL generation rate against gpt-4o's 60.8%. By addressing the scalability, flexibility, and criterion- traceability gaps in existing systems, our framework meets the pressing need for automated, auditable tools that accelerate hospital reporting while maintaining compatibility with layered warehouse standards.

4.4. Implications for Hospital Management

The measurements above were taken to characterise a generation pipeline, but the decisions they bear on are managerial, and the translation is not automatic. Four of them are worth stating in the terms an information-management office would use.

First, the abstention path is an escalation route, not a safety guarantee. Of 791 generated queries, 132 (16.7%) raised a TO-CONFIRM item and 104 of those (78.8%) were warranted, which means that in most cases where the pipeline declared a criterion contested, an indicator owner was in fact the right person to decide. The complementary figure is the one that governs how the output must be handled: recall was 63.0%, so 61 genuinely

ambiguous requests were resolved without asking, and in 42 of those the criterion chosen was wrong. Abstention therefore reduces the volume of silent criterion decisions rather than eliminating it, and a reporting workflow built on this pipeline still needs certification before an indicator leaves the institution.

Second, the ablation identifies which governance asset earns the accuracy. Adding the schema to the prompt raised the effective SQL rate from 22.5% to 40.8% and nearly eliminated nonexistent-field errors, but it did not touch wrong criteria; those halved only when metadata retrieval and historical SQL were added, at 60.8%, and verification carried the rate to 71.7%. The artifact that buys criterion correctness is thus the curated metadata knowledge base and the archive of certified queries—both of which are maintained by people, on a budget, as an ongoing institutional commitment rather than a one-off build. The ablation gives that commitment a measured return, which is the form in which it can be argued for.

Third, the cost figures relocate the bottleneck rather than removing it. Manual authoring takes about 1.8 hours of engineer time per task, and at 650 requests per month the queue is bounded by engineer-hours; a first draft in under two minutes removes that bound. What it does not remove is the need for someone with authority over the indicator to adjudicate the criterion, and Table 8 shows every management use of the output resting on exactly that step. Raising throughput without also raising adjudication capacity converts a visible backlog into an invisible error rate.

Fourth, one of our failures was a data-governance finding rather than a model finding. The unplanned-reoperation query retrieved nothing (Jaccard 0.00) because the model selected a planning flag that is structurally valid, passes schema validation, and is unpopulated across the warehouse. No amount of prompting repairs that; what it identifies is a field the institution believes it collects and does not. A pipeline that grounds generation in metadata will keep surfacing such gaps, and treating them as defects in the data-governance process is more useful than treating them as defects in the generator.

Table 8 sets out the management uses the present results support, each against the reliability actually measured for it and the residual risk that stays with the manager. Every figure in the table is taken from Tables 1–7; the table adds no new measurement.

Table 8. Management uses supported by the present results. Each row pairs a decision the output feeds with the reliability measured for it in this study and the risk that is not transferred to the system. No figure in this table is new; all are drawn from the Results.

Management use	What the pipeline supplies	Reliability measured here	Residual risk retained by the manager
Departmental performance review	Indicator query with the source field recorded for every criterion	Term mapping 52.3% overall, 70.7% in surgery and anesthesia, 34.5% in laboratory	Wrong criteria are 34.7% of all errors and remain executable, so a reported number can be wrong without the query failing
Quality-control monitoring over time	Reusable query text, so the same criterion is applied in successive periods	Effective SQL rate 71.7% (95% CI 63.0–79.0) for the on-premises model	Each cell was run once under deterministic decoding, so run-to-run variability is unestimated
Regulatory and external reporting	Explicit field provenance for audit of how an indicator was computed	Agreement with engineer-certified SQL was high for single-criterion indicators; Jaccard 0.00 and 0.08 for two compound ones	Compound institution-specific definitions still require engineer certification before external release
Criterion governance and catalog upkeep	TO-CONFIRM referrals routed to the indicator owner and recorded	132 of 791 raised (16.7%), 78.8% warranted, recall 63.0%	61 ambiguous requests were not referred, and 42 of those took the wrong criterion
Analyst capacity planning	First draft in under 2 min against about 1.8 h of engineer authoring	About US \$117 per month at 650 requests for gpt-4o, about US \$8 for deepseek-v3, none for the on-premises model	Savings are realised only if criterion adjudication capacity grows with throughput

4.5. Broader Healthcare and Societal Impact728

Beyond the technical result, automated SQL generation changes how often an institution can afford to ask a question of its own data, which is the broader case made for retrieval-grounded language models in health care [52]. Traditional statistical reporting often requires days of engineer time and several rounds of criterion clarification. Our system reduces this to minutes for the first draft, enabling rapid iteration and refinement of indicator definitions with the requesting department in the room. This acceleration could significantly shorten the loop between a quality problem being noticed and being measured, which is a precondition for improvement. Criterion traceability considerations are particularly important given that hospital indicators are reported externally and compared across institutions. Automated generation with explicit field provenance reduces silent criterion drift between analysts and enables systematic auditing of how an indicator was computed, rather than relying on the memory of whoever wrote the query last quarter. The system can actively surface criterion ambiguity as a TO-CONFIRM item and route it to the domain owner for a decision that is then recorded.

Smaller hospitals serving large populations without a data engineering team are where this would matter most; the barrier there is expertise, not infrastructure. We release the synthetic warehouse generator and the validation pipeline for that reason. Beyond it, federated assessment of cross-institutional indicators, generation during morbidity and mortality review rather than through a reporting queue, and a closed loop from indicator definition to query to audit are all plausible extensions. None is tested here.

4.6. Limitations and Future Directions749

The study covers three subject areas (surgery and anesthesia, blood transfusion, laboratory) and therefore says nothing about pathology, physical examination, or insurance-audit reporting, where findings are free text, reconciliation crosses systems, and regulator-defined

indicators change between cycles. The system also relies on the LLM for both extraction and generation, and on ambiguous or poorly specified requests it produced incomplete or overgeneralized queries. Output is also sensitive to prompt design and context length. Requests carrying vague or clinically interpreted terms (“major surgery”, “recent infection”) are the hardest: the system overgeneralizes them or drops a qualifier such as the time window, and these appear in our error analysis mainly as wrong criterion assignments and placeholder insertions. A governed indicator catalog or a rule-based disambiguation step would likely be needed to fix this. The evaluation design is not ours either, and the boundary matters for reading the contribution: the factorial layout, the five prompting strategies, the development-and-validation task split, and the severity ordering of the hallucination categories are adopted from [16], which means this study does not establish that these are the right instruments—it inherits that question. What is ours is the setting they are applied to, the criterion-error category that setting makes necessary, the abstention measurement, and the component ablation. Two reporting gaps remain, and the completed MI-CLAIM and MI-CLAIM-GEN checklists in the Supplementary Materials record each against the item it fails. The two expert raters were not blinded to which model produced a given query, so their scoring could in principle have been influenced by model identity. Each model-prompt-task cell was executed once under deterministic decoding, which fixes the output for a given input but leaves run-to-run variability unestimated; the effect sizes reported here should be read with that in mind. The abstention path and the absence of an external reference point, both of which limited earlier versions of this work, are addressed in Sections 3.9 and 3.11 respectively, though the benchmark figure comes from a single public split and the abstention reference standard rests on the judgement of the same two engineers who built the criterion dictionary.

From an evaluation perspective, although this study incorporated expert-reviewed mapping and structural validation, comprehensive assessment of generalizability across institutions was not conducted; the warehouse schema and the counting criteria are institution-specific by construction, which is precisely the condition that motivates the work and also the condition that limits its external validity. Moreover, while comparison against engineer-certified reference SQL was informative, the small number of included tasks and criteria limits the statistical robustness of the findings. Result-set validation rests on four criteria drawn from three tasks, two of which share a task. No confidence interval on the Jaccard values would be meaningful at that size, and the apparent contrast between simple and compound indicators is a case-series observation rather than an estimate. Future work should focus on extending the applicability of the system to a broader range of subject areas. To improve the coverage and specificity of criterion mapping, the integration of a versioned indicator dictionary and externally curated code sets should be considered. In addition, although cloud models showed high generation rates, their practical use in hospital settings is constrained by the requirement that patient data remain on premises, restricting them to de-identified metadata. Real-time deployment would likely rely on the on-premises models, whose accuracy in this study was at least comparable. Future research should explore more cost-efficient deployment options, such as model distillation or hybrid architectures that route criterion-sensitive sub-tasks to a rule-based component. Finally, large-scale validation studies involving multiple institutions and warehouse schemas will be essential for demonstrating the system’s generalizability and real-world utility. In future work, we plan to compare the performance of alternative models for each subtask, including information extraction, criterion mapping, and SQL generation, to assess their relative accuracy, efficiency, and cost-effectiveness.

5. Conclusions

This study proposed an end-to-end framework that automates the transformation of free-text hospital statistical requests into executable Presto SQL queries over a three-layer clinical data warehouse. Our evaluation revealed unexpected findings: the on-premises qwen2.5:7b model achieved a superior effective SQL generation rate (71.7%) compared to gpt-4o (60.8%), primarily due to better hallucination control. While the framework demonstrated feasibility with 52.3% mapping accuracy, critical challenges emerged in result-set validation, including complete failure on an indicator whose defining flag is unpopulated and minimal overlap where an institutional definition is compound. These findings suggest that while LLM-based automation shows promise, hybrid approaches combining LLM capabilities with rule-based, governed criterion definitions may be necessary for handling criterion-sensitive hospital indicators. For hospital management the operative conclusion is narrower than the headline rate: the pipeline moves the constraint on reporting from engineer-hours to criterion adjudication, and the measured abstention recall of 63.0% means that authority over indicator definitions has to be exercised, not delegated to the generator. Section 4.4 states which management uses the present evidence supports and which risk stays with the institution.

Supplementary Materials: The following supporting information can be downloaded at [\[supplementary URL\]](#): Supplementary Methods, including Figure S1 (detailed pipeline architecture), Table S1 (task-set composition) and Table S2 (integer basis for all model-level rates); Prompt Templates, giving all six pipeline prompts and the five prompting strategies verbatim; SQL Evaluation Metrics, listing the 72 expert-rating items; and the completed MI-CLAIM and MI-CLAIM-GEN checklists.

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Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of West China Hospital, Sichuan University (protocol code [\[XXXX-XXXX\]](#), approved 27 July 2026).

Informed Consent Statement: Patient consent was waived because the study used only de-identified operational metadata and aggregate statistical query results, with no intervention and no access to identifiable patient records.

Data Availability Statement: The source code for the pipeline and the hallucination detector, together with the SynHDW generator and its schema manifest, is available from the corresponding author upon reasonable request, subject to the data-governance rules of West China Hospital, Sichuan University. The de-identified metadata schema, the metric-criterion dictionary and the evaluation task set are available under the same conditions. Patient-level warehouse data cannot be shared.

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Conflicts of Interest: A related invention by the author (patent AJ2534335, a large language model– and reinforcement learning–based natural language to structured SQL generation and optimization system) is declared as prior disclosure. The present work uses retrieval augmentation and execution feedback *without* reinforcement learning. The author declares no other conflict of interest. The funders had no role in the design of the study; in the collection, analyses or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

Use of Generative Artificial Intelligence

Generative AI tools were used during the preparation of this manuscript for language editing and for drafting sections of the text, which the author subsequently reviewed and revised. No text, data, figure or citation was accepted without author verification, and the author takes full responsibility for the content of the publication. The large language models studied in this work are the object of the research and are described in Section 2.6; their use as study subjects is distinct from the manuscript-preparation use disclosed here.

Abbreviations

ASA: American Society of Anesthesiologists
CDW: clinical data warehouse
DC: data center (warehouse layer)
DL: data lake (warehouse layer)
HSD: honestly significant difference
IRB: institutional review board
LLM: large language model
MDR: master data repository (warehouse layer)
PACU: postanesthesia care unit
RBFM: rule-based field matcher
SQL: Structured Query Language
SynHDW: Synthetic Hospital Data Warehouse

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